Rajasekhar Reddy Kolagotla

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Professional Summary

Backend Engineer with **5+ years** building scalable **Java-based microservices and distributed systems** for **real-time financial platforms** and **risk management systems**. Experienced in designing **high-throughput backend services** using **Spring Boot**, **Kafka**, and **cloud-native** infrastructure on **AWS**. Strong background in **event-driven** architectures, **transactional** systems, and **database performance optimization**. Hands-on experience building AI-powered backend services using **RAG pipelines** and **vector search** for real-time analytics. I specialize in owning **backend features end-to-end**, turning complex **business workflows** into **production-ready services** that **scale**.

Technical Skills

- **Programming Languages:** Java, Python, JavaScript, TypeScript
- **Backend & Frameworks:** Spring Boot, Spring MVC, Spring Security, Hibernate, Microservices Architecture, REST APIs, JSON
- **Databases & Data Stores:** MongoDB, PostgreSQL, MySQL, SQL Server, DynamoDB, Cassandra
- **Cloud Platforms:** AWS (ECS, EKS, EC2, S3, Lambda, CloudWatch), Azure, GCP
- **Distributed Systems & Messaging:** Apache Kafka, Spring Kafka API, RabbitMQ, Event-Driven Architecture, Asynchronous Processing
- **Containerization & Orchestration:** Docker, Kubernetes
- **DevOps & CI/CD Tools:** Jenkins, Git, GitHub, Bitbucket, Maven, Gradle, Terraform, CI/CD Pipelines, Code Reviews
- **Testing & Documentation:** JUnit, Mockito, Postman, Swagger, SonarQube
- **Monitoring & Logging:** Amazon CloudWatch, Splunk, New Relic, log4j
- **Frontend Technologies:** React, Angular, HTML5, CSS3, Bootstrap, Tailwind
- **AI & Data Systems:** Retrieval Augmented Generation (RAG), LLMs, LangChain, Vector Databases, Embedding Models

Work Experience

Software Engineer **Feb. 2024 - Present**
Wells Fargo | San Francisco, CA | Contract

- **Led the redesign of Java 17 Spring Boot microservices** supporting ACH, wire, and peer-to-peer payments, improving transaction processing efficiency and reducing service bottlenecks.
- **Developed backend services** supporting **high-volume** financial transactions across distributed payment systems.
- **Standardized REST API contracts** across distributed payment services, ensuring backward compatibility and reducing integration inconsistencies between microservices.
- **Optimized MongoDB data models and aggregation pipelines**, implementing compound indexes and configuring replica sets to support high availability and reliable data replication.
- **Designed and tuned PostgreSQL schemas and queries** supporting transactional datasets used in risk evaluation and reporting systems.
- **Implemented secure REST APIs** using Spring Security with JWT and OAuth 2.0, enforcing role-based access control and API security policies aligned with OpenAPI standards.
- **Built Kafka-based event pipelines** to decouple payment processing from downstream risk and reporting workflows, enabling asynchronous communication between services.
- **Containerized backend services using Docker** and deployed them to AWS ECS/EKS environments with autoscaling, health checks, and rolling deployments.
- **Developed CI/CD pipelines** using Jenkins and GitHub Actions integrating automated testing, Docker image builds, and Terraform-based infrastructure provisioning.
- **Improved system observability** by implementing structured logging, CloudWatch monitoring, and Splunk-based diagnostics for production services.
- **Collaborated on backend architecture and code quality practices** through design discussions, peer code reviews, and Agile development workflows.

Software Engineer **Jan. 2019 - May 2022**
TCS | Kolkata, India | Full-Time

- **Developed Spring Boot microservices** supporting customer onboarding, KYC validation, account creation, and loan approval workflows for banking platforms.
- **Built REST APIs** enabling communication between internal banking services and third-party financial integrations using API-first development practices and Swagger documentation.
- **Implemented Apache Kafka event pipelines** to support asynchronous processing for notifications, document verification, and backend workflow coordination.
- **Designed and optimized relational and NoSQL database schemas** across PostgreSQL, MySQL, and MongoDB to support high-volume transactional data processing.
- **Containerized backend services using Docker** and deployed them to Azure Kubernetes clusters to support scalable and fault-tolerant cloud deployments.
- **Contributed to Jenkins CI/CD pipelines** to automate build, testing, and deployment workflows for microservices.
- **Mentored junior developers** on Spring Boot development, debugging techniques, and REST API design practices.
- **Collaborated with QA, DevOps, and product teams** to deliver backend features and provide production support for critical financial systems.

Personal Projects:

Real-Time Payment Fraud Detection Platform

- **Designed and implemented a real-time fraud detection backend** that processes financial transactions and identifies suspicious activity using rule-based and anomaly detection techniques.
- **Built event-driven transaction pipelines using Apache Kafka** to enable streaming analysis of payment events and asynchronous fraud evaluation.
- **Developed backend services using Spring Boot** to evaluate transaction patterns, generate fraud alerts, and expose investigation APIs.
- **Created a monitoring dashboard and APIs** to visualize fraud alerts, transaction activity, and risk indicators for investigation workflows.
- **Containerized the platform with Docker and deployed it on AWS**, enabling scalable event processing and real-time analytics.

AI Analytics Platform for Drone Telemetry (RAG Microservice)

- **Designed and built a backend analytics service** that ingests high-volume drone telemetry data and exposes analytics through REST APIs for operational insights.
- **Implemented a Retrieval Augmented Generation (RAG) pipeline** using LangChain and embedding models to enable intelligent query responses over telemetry logs.
- **Developed vector similarity search workflows** to support low-latency querying and pattern detection across large telemetry datasets.
- **Containerized the service using Docker** and deployed it on AWS infrastructure to support scalable processing and API-based access to analytics.

Education:

M.S. in Computer Science - California State University, Sacramento **Aug. 2022 - Dec. 2024**
B.S. in Computer Science - KL University, India **Aug. 2014 - June 2018**

Certifications:

- **AWS Certified Solutions Architect - Associate**
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**
- **Google Prompting Essentials**
- **Building RAG Agents with LLMs (Nvidia)**